



Molecular and Physiological Adaptations to Endurance Training

3

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Introduction

The purpose of this chapter is to provide an overview of the research addressing the molecular and physiological adaptations to endurance exercise training. The history of research examining the adaptive responses of the human organism to endurance exercise training is lengthy. One can argue the historical reports of the physical training by Greek Olympians or Roman military units in ancient civilizations may be some of the first recorded information. More contemporary reports based upon the use of the “scientific method” and empirical evidence begin in the eighteenth century and span to the present day [1]. The progression and details of this latter work over the last four centuries is fascinating and provide a contextual framework for the understanding of the scientific investigations done in our present, contemporary times. The historical context and insights of this prior research provides the organization constructs of the topics covered in this chapter. Specifically, these topics are

- Molecular adaptations
- Cardiovascular adaptations
- Metabolic adaptations
- Skeletal muscle, tendon, and bone adaptations
- Hormonal-endocrine adaptations

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Each one of these topics warrants and deserves a chapter unto themselves, as a simple PubMed and Scopus search on these topics results in nearly 100,000 publications suggesting there is ample evidence to discuss; but, space limitations herein necessitate addressing these topics in a concise and succinct fashion only. The reader is directed to some of the key literature cited within this chapter if they desire more extensive discussions on elements of any of the select topics.

Adaptive Responses to Endurance Exercise Training

Molecular Adaptations

Molecular mechanisms are critical in allowing organisms to adapt to and survive diverse environmental challenges—whether imposed or selected by the organism—hence understanding these mechanism provides great insight to the physiological capacity. In recent years, genomic and proteomic approaches have been key drivers of advancement in the field of biological sciences, e.g., providing knowledge about gene and protein expression, regulation of signal transduction pathways, and functional control of enzymes/proteins by reversible protein phosphorylation. These scientific advances have been evident in exercise physiology and sports performance as well as other of the biological life sciences.

Research evidence strongly supports that the repetitive muscular contractions with endurance exercise training generates specific mechanic stimuli which result in and promote adaptation. That is, there are a series of primary mechano-sensor stimuli “switching on” this adaptive response at the molecular level. Some of the key stimuli involved with this mechanism include muscular stretch, intracellular calcium flux, fuel substrate utilization, overall stored energy status, and oxygen stress within the organism [2–4]. Furthermore, it is important to recognize the necessity for exercise training to be performed at appropriate levels (e.g., intensity, duration, frequency) in order to invoke stimuli-driven perturbations of a magnitude to induce adaptation.

The molecular adaptation response(s) invoked by these exercise stimuli center on taking a sequence of the genetic code from DNA for a specific cell protein and generating functional gene products—termed gene expression—for critical endurance-related physiological processes such as mitochondrial respiration and biogenesis, signaling and catalytic enzymes, and transporter proteins. To this end, it is well-recognized exercise-induced increase in skeletal muscle mitochondrial content—specifically termed mitochondrial biogenesis—is fundamental to endurance training adaptation and a key regulator of this process is the transcription co-activator PGC-1 α (peroxisome proliferator-activated receptor gamma co-activator 1-alpha). Figure 3.1 depicts the basic aspects of the PGC-1 α pathway through which this process is activated with exercise training and the critical outcomes from activation [4, 5].

Additionally, endurance training mediates molecular-based changes in substrate availability and utilization. For example, the capacity to transport glucose and fatty acids into and from the blood into the muscle cells are enhanced by increasing the

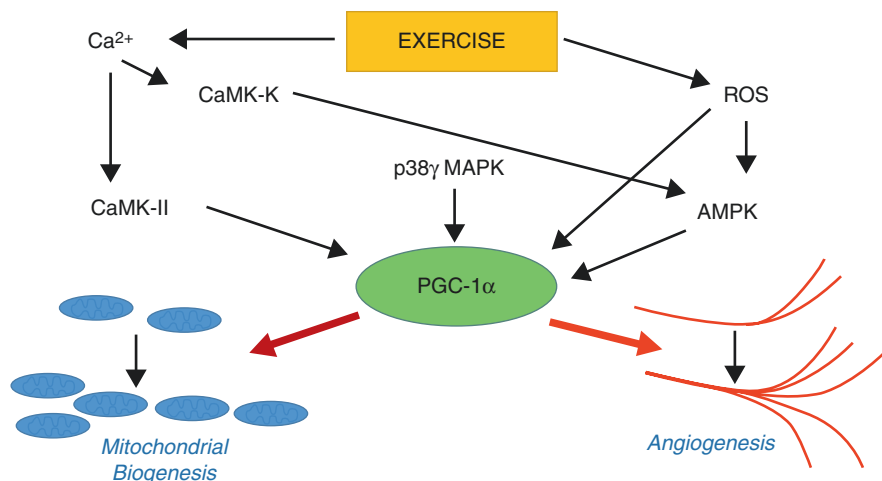


Fig. 3.1 Proposed signaling pathways involved in exercise-induced peroxisome-proliferator-activated receptor- γ co-activator-1 α (PGC-1 α) regulation in skeletal muscle. Current evidence suggests roles for calcium (Ca^{2+}), calcineurin (CnA), Ca^{2+} /calmodulin-dependent protein kinases (CaMK), AMP-activated protein kinase (AMPK), and p38 γ mitogen-activated kinase (p38 γ MAPK) in PGC-1 α regulation. ROS reactive oxygen species

level of the substrate transporters GLUT4 and FAT/CD36. Glucose transporter type 4 (GLUT4) is critical in their metabolic role as an insulin-regulated facilitator of glucose uptake in a resting state and their ability to be insulin independent during exercise. FAT/CD36 (fatty acid translocase) is equally important as a membrane transporter increasing the uptake of fatty acids into the muscle for use in lipolysis-based metabolism-respiration (i.e., β -oxidation). As mitochondrial biogenesis is augmented, there is an enhanced capacity for β -oxidation and the Citric Acid Cycle (CAC; frequently referred to as the Krebs Cycle) ATP production. This leads to a greater potential utilization of these key substrates during exercise as uptake capacity increases (see later section—Metabolic Adaptation) [4, 6].

A key extra-cellular response is angiogenesis, the physiological process through which new blood vessels are formed. This development results in an improved capillary to fiber ratio at the skeletal muscle, leading to an enhanced blood flow and oxygen delivery to the muscle (see next section). The vascular endothelial growth factor (VEGF) protein has been identified as a central regulator of angiogenesis that is released by the skeletal muscle to stimulate vascular growth. Interestingly, the expression of VEGF in muscle appears to be one of the genes whose expression is also regulated by the PGC-1 α pathway (Fig. 3.1) [4, 6, 7].

The maximal activation of the molecular adaptation response and the increase in the functional proteins in skeletal muscle by exercise training is rapid. In fact, this is achieved within hours to days of repeated endurance exercise exposure (Fig. 3.2). In time, however, with the repeated exposure the stimulus is deemed maximal and plateaus (shortly thereafter) as exercise training exposure continues (i.e., unless there is a progression to higher levels of stimulus—“progressive overload”) [4].

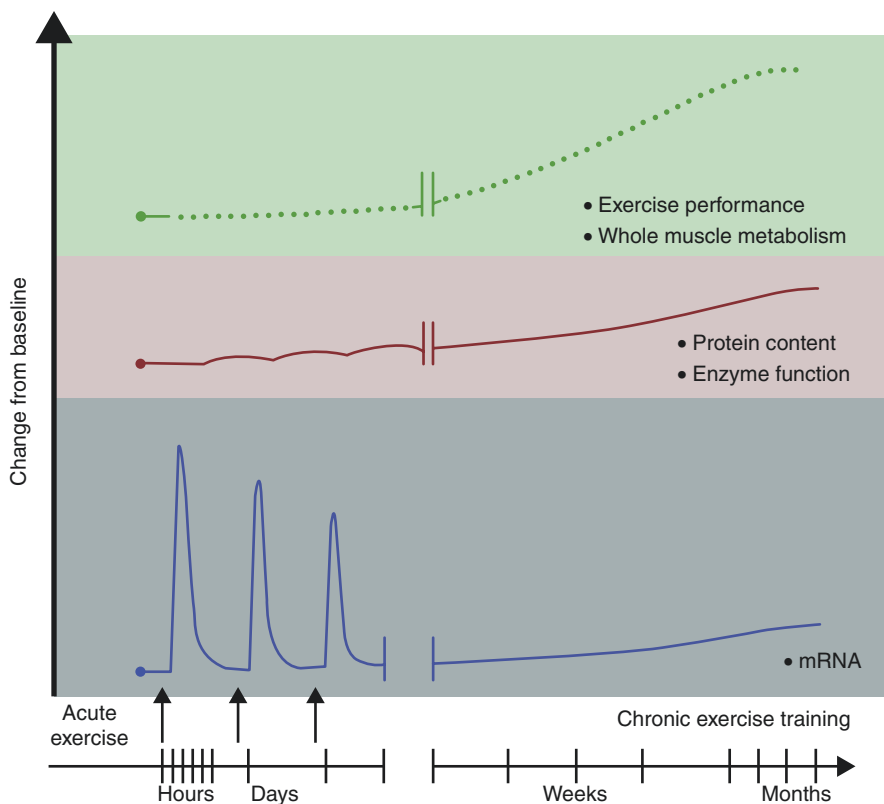


Fig. 3.2 Diagram depicting changes in gene (mRNA) expression (bottom panel) and protein content (middle panel) over time as a consequence of acute exercise and chronic (repetitive) exercise training. With gene upregulated by exercise and training, an individual exercise session elicits a rapid, but transient, increase in relative mRNA expression of a given gene during recovery. Alterations in mRNA expression are typically greatest at 3–12 h after cessation of exercise and return to basal levels within 24 h. Translational processing and an elevated rate of post-exercise protein synthesis result in the same-directional change in protein content. Chronic exercise session exposure results in the gradual accumulation of protein in response to repeated, pulsed increases in relative mRNA expression. Long-term adaptation to training is due to the cumulative effects of each acute exercise session leading to a new functional threshold. Training-induced changes in protein content/enzyme function lead to altered metabolic responses to exercise (e.g., substrate metabolism), resulting in improved exercise performance (upper panel). Used with permission; Egan B, Zierath JR. Exercise metabolism and molecular regulation of skeletal muscle adaptation. *Cell Metabolism*. 2013; 17(2): 162–184

Cardiovascular Adaptations

Cardiovascular response to exercise revolves around the classic physiological elements of heart rate, stroke volume, cardiac output, mean arterial pressure, and total peripheral resistance. These determinants of cardiovascular function influence the primary physiological factors associated with success in endurance exercise events: maximal oxygen uptake ($\text{VO}_{2\text{max}}$), the ability to work at a high fractional utilization of $\text{VO}_{2\text{max}}$, and overall work economy (*N.B.*, the terms “economy” and “efficiency”

are sometimes used synonymously. However, economy refers to the relationship between oxygen consumption and movement speed [e.g., running velocity], while efficiency is the ratio between the mechanical energy produced during exercise and the energy cost of the exercise.).

These three factors associated with endurance exercise success are influenced by cardiovascular function through its central role in oxygen delivery—a well-documented key determinant of aerobic capacity and performance in endurance athletes [3]. An efficacious endurance exercise training program will result in an athlete developing an augmented $\text{VO}_{2\text{max}}$, the ability to work at a higher fractional utilization of $\text{VO}_{2\text{max}}$, and an improved work economy [3, 8].

Enhancement of cardiac output plays a central role in the determination of the improvement in $\text{VO}_{2\text{max}}$. While $\text{VO}_{2\text{max}}$ has a strong heredity component to potential genotypic expression, the ability to facilitate the cardiac output (CO) through training is a powerful effector of $\text{VO}_{2\text{max}}$ phenotype expression as maximal CO can be 20–50% greater in endurance-trained versus untrained, matched individuals [8–10].

The major cardiac adaptations to improve CO from endurance training is the enhanced capacity for ventricular filling ($\uparrow$ end-diastolic volume [EDV] parameters) and the ability to use the Frank-Starling mechanism during exercise to increase the ejection fraction. Collectively, these changes result in an improved capacity to increase stroke volume. Key factors responsible for increased EDV capacity after training include enhancement in left ventricle (LV) compliance, increased cardiac dimensions (i.e., pre-load LV hypertrophy), increased rate of LV pressure decline, reduced pericardium-mediated diastolic ventricular interactions, enhanced diastolic suction, increased rate of calcium uptake within myocardial sarcoplasmic reticulum, and increased changes in vascular volume. Taken together, these changes result in a greatly improved pumping capacity (CO) for the heart [8–10]. Also, coupled to these changes are training-induced blood hypervolemia and erythropoiesis leading to greater hemoglobin content and, therefore, oxygen carrying capacity and arterial-venous oxygen differential (a-v O_2 difference) [3, 8, 9]. In short, essentially all those factors driving the $\text{VO}_{2\text{max}}$ capacity changes as determined by the Fick equations are enhanced (see following equation [3]).

$$\text{Fick Equation : } \text{VO}_2 = \text{CO} \times \text{a-v } \text{O}_2 \text{ difference}$$

Relative to the lactate threshold, endurance exercise training results in a shifting of the “break point” for accelerated anaerobic energy contributions during an exercise session to a higher exercise intensity [2]. Such representations are typically depicted and quantified by examination of the lactate threshold profile responses in athletes [3]. This occurrence is driven to a large extent by the increases in mitochondrial biogenesis (noted earlier) leading to a greater mitochondrial density. This biogenesis coupled with the improved oxygen delivery capacity of the cardiovascular system leads to reduced lactate production and accumulation due to increased pyruvate oxidation. Metabolically, this adaptation also means increased ability of muscle cells to uptake and oxidize fat and reduce carbohydrate oxidation through glycolysis (i.e., reduced pyruvate formation, see the following section). Furthermore, the increased CO ability following training translates into increased blood lactate removal during exercise [2, 3]. These events allowing for a shift of the lactate

Table 3.1 Summary of the cardiovascular adaptation resulting from an exercise training program (i.e., specifically one focusing upon aerobic endurance-based physical activities)

Measurement	Resting	Submaximal exercise	Maximal exercise
Heart rate	↓	↓	↓, NC
Stroke volume	↑	↑	↑
Cardiac output	↓, NC	↓, NC	↑
a-v O ₂ diff	↑, NC	↑	↑
VO ₂	NC	↓, NC	↑
Systolic BP	↓, NC	↓, NC	NC
Diastolic BP	↓, NC	↓, NC	↓, NC
MAP	↓, NC	↓, NC	NC
TPR	NC	↓, NC	NC
Blood volume	↑	–	–
Plasma volume	↑	–	–
Erythrocyte mass	↑	–	–
Heart volume	↑	–	–

Symbols: ↑ increase, ↓ decrease, *NC* no change, – not applicable

Abbreviations: *a-v O₂ diff* arterial-venous oxygen difference, *VO₂* oxygen uptake, *BP* blood pressure, *MAP* mean arterial pressure, *TPR* total peripheral resistance

breakpoint to a higher intensity helps the athlete to perform at higher intense efforts with reduced likelihood of muscular fatigue [2]. Finally, the enhanced metabolic capacity-efficiency from training combined with an improved biomechanical efficiency through adaptation to chronic repetitive movement patterns yields an improved exercise economy (i.e., less energy required for any given level of movement) [3, 8].

Table 3.1 summarizes the adaptive responses in the critical cardiovascular components in response to an endurance exercise training program [2, 3, 8, 9].

Metabolic Adaptations

This section focuses on the metabolic changes within skeletal muscle that are associated with the improved capacity and performance from endurance exercise training. In general, the key metabolic adaptive component of endurance training is the increased capacity for skeletal muscle to oxidize fuel for energy production. In light of the fact that a marque endurance event such as a running marathon (42.2 km) can have a total energy cost greater than 10,000 kJ in a roughly 2 h period, an increased capacity and rate for fuel oxidation is critical to improving performance (*N.B.*, long course triathlons, cycling races, or certain Nordic ski events can exceed even this value for energy cost!) [2, 3].

The increased fuel utilization capacity is a direct result of increased mitochondrial content (see earlier biogenesis discussion) and the associated enzyme activity profile from that content change; in particular, those enzymes associated with β -oxidation and the CAC energy pathways of aerobic metabolism [2]. Essentially an increased metabolic flux of both energy pathways results in a greater production of electrons for the electron transport chain and, hence, enhanced and ultimately upregulated

mitochondrial oxidative phosphorylation (aerobic metabolism) (*N.B.*, metabolic flux is the rate of turnover of molecules through a metabolic pathway). The up-regulation of oxidative phosphorylation enables a more efficient energy production and reduces the glycolytic pathway flux and therefore the accumulation of cytosolic waste products that contribute to muscle fatigue associated with that pathway [11–13].

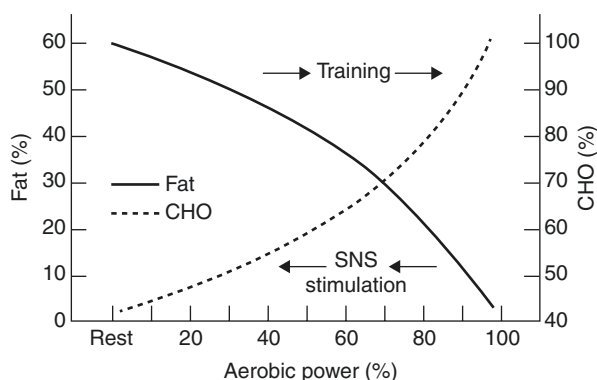
Additional consequences of the metabolic adaptation to training are an increased utilization of fats as fuel and a subsequent decrease in the utilization of carbohydrate as a fuel at any given submaximal exercise intensity. This shifting of fuel use results in a muscle glycogen sparing, thereby attenuating muscle glycogen use and depletion during prolonged submaximal exercise and abating another potential fatigue factor (i.e., muscle glycogen depletion is a critical factor in an endurance event of sufficient duration [~ 60 – 90 min, $>60\%$ $\text{VO}_{2\text{max}}$]). Along these lines, following training, there is an increased level of resting muscle glycogen stores and intramuscular triglycerides, resulting in a more efficient use of fuels due to proximity to the exercising muscle tissue. Additionally, the availability of extra-muscular fuel sources is also enhanced due to transporter changes noted and discussed earlier in the Molecular Adaptations section [11–14].

In brief, endurance exercise training shifts the metabolic energy production towards a greater reliance on fats as a fuel, leading to a stored carbohydrate sparing. Figure 3.3 conveys this change in fuel utilization with training, as a shifting in the “cross-over” point to higher exercise intensities. At the same time, these are an enhancement of carbohydrate stores (i.e., glycogen). Coupled with the molecular adaptation noted earlier, this leads to a greatly improved ability in the rate and yield of ATP production during exercise, thereby, increasing the capacity for physical work, overall and at higher intensities.

Skeletal Muscle, Tendon, and Bone Adaptations

The skeletal-musculotendinous unit is a remarkable organ system within the human body and has a tremendous capacity to display plasticity. The positive adaptations in this system due to its plasticity are significant contributors to the performance enhancement associated with endurance training [1, 2, 15].

Fig. 3.3 Cross-over concept and the influence of endurance exercise training. *CHO* carbohydrate, *SNS* sympathetic nervous system. (Image adapted from: Brooks GA, Mercier J. Balance of carbohydrate and lipid utilization during exercise: the “cross-over” concept. *Journal Applied Physiology*. 1994;76 (6): 2253–2261)



In skeletal muscle, the smallest functional unit to be activated (i.e., recruited) and utilized in tension development is the motor unit and the muscle fibers within that unit. Individual motor units have fibers of identical characteristics, but intact whole skeletal muscle is comprised of three distinct and differing fiber types. The nomenclature utilized to distinguish these fiber types have varied since their identification some 50+ years ago, which can lead to confusion when reading literature [2, 15, 16]. Here in the Roman numeral naming system is employed, that being Type I, Type IIa, and Type IIb/x. Type I are “slow-twitch” oxidative fibers which are slow in force generation have a highly developed oxidative capacity due to high mitochondrial content and oxidative enzyme expression and a dense capillary supply. Type IIa are “fast-twitch” oxidative fibers which are relatively fast in force generation but have similar oxidative profiles to Type I fibers. Type IIb/x are “fast-twitch” fibers with a highly developed glycolytic metabolic profile (i.e., rich in glycolytic enzymatic expression), poor mitochondrial and capillary development, and fast force generation capacity. For these fibers, the tension development capacity within twitch responses is $\text{IIb/x} > \text{IIa} > \text{I}$ (i.e., greater to lesser). Research studies where fiber-typing has been conducted supports that athletes in endurance sports tend to have a higher proportion of Type I fibers in the musculature used in their sporting activity [17]. That said, though, the evidence does not support that fiber type alone serves as a strong predictor of performance capacity [15, 17].

Earlier it was noted that endurance training facilitates changes in mitochondrial content, oxidative enzymatic expression, and capillary density within skeletal muscle fibers. These characteristics are components of the classification criteria for muscle fiber-typing. Changes due to training in these characteristics beg the question—“Does exercise training result in fiber type transformation?” This has long been a question in exercise research, with some contradictory findings when a variety of animal species were examined [1, 15]. Currently, there is no direct experimental evidence for fiber-type transformation in humans. It is important to remember that the fiber type determined is a function of a multitude of neural, metabolic, functional, histological, and morphological characteristics [3]. Nonetheless, chronic endurance training does transform the biochemical characteristic phenotype of muscle fibers towards the Type I metabolic profile, but the extent of such expressional change varies with individual fiber types [1, 16]. Moreover, there is at least some evidence that regular endurance training (cycling) may also induce muscle hypertrophy in previously untrained subjects [18].

In a related fashion, resistance training is associated with neural and hypertrophic adaptation which leads to improved muscular strength-power capacities. For years, it was postulated resistance training and endurance training interfered with the adaptive responses of each individual form of training. Recent work, however, supports when endurance and resistance training are applied concurrently there can be some degree of an optimization of the magnitude of muscle hypertrophy, maximal strength, and endurance capacity development [18]. These effects have been observed in untrained as well as trained runners and translate into improved high-intensity running characteristics with obvious implications to performance [19].

Nonetheless, this line of research work is not entirely conclusive and is in need of much further exploration.

Tendon and bone are instrumental to the health and functionality of the endurance athlete, yet relative to some of the other adaptive components discussed in research literature they are seldom addressed as extensively [20–22]. Frequently, the available related discussions focus upon injury, damage, or maladaptation with health-related comorbidities such as the osteoarthritis or the female athlete triad [22–24].

Tendons display good ability to adapt to different states of loading through movements encountered in endurance activities; that is, provided the timeframe for adaptation is sufficiently long enough and gradual in loading (e.g., weeks, months, and years) [20] although the long-term effect of chronic exercise training on tendon health status over a lifetime seems inadequately addressed in research. Similar to a tendon, bone shows comparable positive responsiveness to endurance exercise, as regular and varied loading has constructive effects on bone health [21]. Consistent exercise loading consisting of high-magnitude impacts is highly stimulatory to osteogenic activity of the bone promoting thick cortical bone development. Furthermore, endurance training is a good overall promoter for improvement of bone strength through adaptation in geometric modeling. However, not all endurance training activities are universal in this effect. For example, running is associated with increased bone density, particularly in the legs, whereas cycling is associated with a mild decrease in bone density in the spine; and, in athletes who do both, running exerts a stronger influence than cycling [25]. These effect differences being most likely do to weight bearing vs. non-weight bearing aspects of the activities.

Collectively, these tendon-bone adaptive responses to endurance training result in an improved resilience to injury from the chronic, repetitive nature of the exercise activity although chronic over-use or overtraining exposure are counterproductive and have negative effects and greatly increase the risk of injury (see Hormonal-Endocrine Adaptation section) [20–26].

Hormonal-Endocrine Adaptations

In humans, there are a variety of hormones and hormone-acting substances that exhibit endocrine, paracrine, and autocrine actions on a multitude of tissues. These substances are involved in the regulation of many physiological processes such as metabolism, growth and development, hydration, cardiovascular functions, immune responses, and stress reactivity. A large number of research findings support that endurance exercise training profoundly affects hormones, the endocrine glands/tissues responsible for their production, and thus the processes these substances regulate and control [2, 3].

Acute exercise is a powerful stimulant to the endocrine system and results in a multitude of hormonal changes. Table 3.2 summarizes the effects of a single acute, endurance exercise session (of various intensity, duration) on the major human

hormones; *in nuce*, essentially exercise serves as a stressor provoking increases in circulating levels of the hormones (exception → insulin↓)—the more demanding the exercise stress, the greater the response. Table 3.2 also depicts how endurance training (chronic) effects hormones response to subsequent exercise sessions—i.e., generalizing, the training responses is to lower the blood level of hormones at rest, and in response to performing an acute exercise session post-training [26–29]. The greater the overall training adaption incurred typically the greater the attenuation of the hormonal response to exercise. The exception to this is maximal or supramaximal exercise. In these situations, the training adaptations allow for a greater level of workload to be performed in maximal or supramaximal efforts and the greater workload achieved produces a greater physiological stimulus and thus commonly a greater hormonal response [26, 27].

In most situations, when endurance exercise training regimes are well constructed and executed, the hormonal effects (adjustments) are highly positive and lead to improved exercise capacity and through genomic and non-genomic means facilitate many of the adaptation processes already noted in this chapter [26–28]. Regrettably, however, poorly developed or implemented exercise training regimes can result in counterproductive hormone responses and negative physiological consequences (maladaptations).

The attenuated endocrine responses following endurance training come about by a greater sensitivity of target tissue to the hormonal stimulus and diminished levels of neural, humoral, and hormonal stimuli disturbances in the blood and other body fluids regulating the various endocrine glands [29]. Relative to the former point of sensitivity, in response to an exercise training program, many target tissues increase the expression of functional hormone receptors, receptor affinity for hormones becomes increased, and post-receptor amplification mechanisms in the cells of target tissues are typically increased. Essentially, all these changes result in a target tissue needing less amount of a hormonal signal to bring about a physiological outcome/change in “down-stream” events within cells [29–31].

Hormonal-Endocrine Adaptations

If training regimes are excessive in the amount of intensive training or overall volume of training and, or an athlete has too many additional life stresses compounding their situation while training, it is possible for maladaptive hormone response occurrences. Such responses are usually associated with overreaching or overtraining scenarios [32]. In brief, normally, if an individual is subjected to gradual increases of “training overload” stress followed by a period of time to rest and recovery they will adapt in a positive fashion and promote adaptations leading to an improved performance capacity. If the training overload stress is too much, or adequate rest is not allowed, then the athlete may not be able to adapt and their performance declines rather than improves. That is, they may progress from “normal” training to experiencing “overreaching” training and in due course “overtraining.” If not checked in

Table 3.2 Generalized hormonal responses to “acute” endurance-type exercise of varying intensity and duration

Hormone	Principal physiological function ^a	Exercise response			
		Acute			Chronic
		Short-term submaximal (e.g., <30min; ≈, >LT)	High intensity (e.g., ≥LT)	Prolonged submaximal (e.g., >60 min; <LT)	Endurance training
ACTH	Adrenoregulatory	↑	↑↑	↑↑	↔, ↓
AVP	Hydration, fluids	↑ (>50% VO _{2max})	↑↑	↑↑	↔
Aldosterone	Hydration, fluids	↑	↑↑	↑↑	↔
Catecholamines	Catabolic, (e.g., lipolysis, glycogenolysis), cardio-regulatory	↑	↑↑	↑↑	↔, ↓
Cortisol	Catabolic (e.g., lipolysis, proteolysis), stress reactivity	↑ (>60% VO _{2max})	↑↑	↑↑	↔, ↓
Estradiol-β-17	Bone metabolism, catabolic (e.g., lipolysis), reproductive	↑	↑	↑ ↓ (if excessive)	↔, ↓
FSH-LH	Reproductive	↑	↑, ↔	↑, ↓, ↔	↓
Glucagon	Glucoregulatory	↑	↑	↑↑	↓
Growth hormone	Anabolic (e.g., myoplasticity, osteogenic)	↑	↑↑	↑↑	↔, ↑
Insulin	Glucoregulatory	↓	↓	↓	↓
Parathyroid	Calcium metabolism	↑	↑	↑	↔

(continued)

Table 3.2 (continued)

Hormone	Principal physiological function ^a	Exercise response			
		Acute		Chronic	
		Short-term submaximal (e.g., <30min; ≈, >LT)	High intensity (e.g., ≥LT)	Prolonged submaximal (e.g., >60 min; <LT)	Endurance training
Prolactin	Immune, stress reactivity	↑	↑↑	↑↑	↑, ↓
Progesterone	Reproductive	↑	↑	↑ ↓ (if excessive)	↔, ↓
Testosterone	Anabolic (e.g., myoplasticity, gluconeogenesis, erythropoiesis)	↑	↑	↑ ↓ (if excessive)	↔, ↓
T ₄ -T ₃	Calorigenesis, endo-permissive	↔	↔	↑, ↓	↔, ↓
TSH	Thermoregulatory	↑	↔, ↑	↔, ↑	↔

The influence of “chronic” endurance exercise training on hormonal response to subsequent exercise after training is also depicted [e.g., exercise ACTH still ↑s after training, but the magnitude of the ↑ is either unaffected (↔) or slightly lessened (↓)]
Abbreviations/symbols: *LT* lactate threshold, ↑ increase, ↑↑ strong increase, ↓ decrease, ↔ no change, *ACTH* adrenocorticotrophic hormone, *AVP* arginine vasopressin, *FSH-LH* follicle-stimulating hormone–luteinizing hormone, *T₄-T₃* thyroxine–triiodothyronine, *TSH* thyroid-stimulating hormone

^aPrincipal role relative to exercise

this progression, the athlete may ultimately develop and display the clinical characteristics of the Overtraining Syndrome, a serious medical condition [32, 33].

The research in endurance athletes who are overreaching or overtraining suggests the endocrine system response is in two phases: an initial hyperactivity phase, followed by a latter hypoactivity phase [33, 34]. In the hyperactivity phase, elevations in the circulating levels of hormones such as adrenocorticotrophic hormone, cortisol, prolactin, and catecholamines have been reported at rest and/or in response to an acute exercise session [33, 34]; although, these hormonal findings are not completely universal [32]. This hyperactivity phase may be reflective of the “overreaching” status in the training continuum. Interestingly, it should be noted that in some situations during “overreaching” if the athlete is given short-term rest and recovery (~2 weeks) they may actually compensate with greater than normal adaptations and enhancements of performance (i.e., tapering) [32]. The hypoactivity phase seems to more closely correspond to the classic “overtraining” status and, or with the occurrence of the Overtraining Syndrome medical classification—i.e., a state of chronic fatigue, lethargy, and underperformance. For this phase, certain glucoregulatory and reproductive hormones are found to be suppressed [34, 35]. The development of the hypoactivity phase appears to be the more serious outcome as it may require months of rest and recovery by the athlete in order to regain normal endocrine function and performance capacity [34]. Whether this state and the attenuated hormone responses associated with it represent an endocrine gland exhaustion or regulatory axis dysregulation remains unclear.

The exact physiological mechanism that induces the Overtraining Syndrome is unknown, but for many years it was thought to reflect some degree of the final stage (exhaustion) of Selye’s General Adaptation Syndrome model [36]. However, Dr. Lucille Lakier Smith [37] proposed in 2004 that overtraining causes too much excessive musculoskeletal loading which becomes compounded by inadequate rest, recovery, and poor nutrition resulting in tissue damage-related local and systemic inflammation development. Pro-inflammatory cytokines such as IL-6, TNF- α , and IL-1 β become elevated and act upon multiple levels of the hypothalamic-pituitary-adrenal (HPA) medullary-cortical axis, influencing the hormones of the axis which are associated with impacting effective mood state, sexual function, metabolism, and cardiovascular function [34, 37]. Furthermore, the cytokine changes noted may facilitate a suppression of the cell-mediated aspects of the adaptive immune system. This creates an increased risk for infection and the so-called sick response which is associated with many of the characteristics found in overtrained athletes [38, 39]. This proposed mechanism, while not entirely encompassing of all symptomology, does reconcile and connect many of the major pathogenic and clinical manifestations associated with the Overtraining Syndrome. Nonetheless, more experimental research and empirical data are necessary on this topic.

A relative new terminology associated with exercise and endocrine-related maladaptation is “Relative Energy Deficiency in Sports” (RED-S). This was proposed by an International Olympic Committee Medical Commission working-group as a new language to apply to the reproductive endocrine disorders that affect female and male athletes [40], typically in a high prevalence within endurance-related sports.

Principally, these disorders are the “Female Athlete Triad” and the “Exercise-Hypogonadal Male Condition” (EHMC) [41, 42]. RED-S was proposed as an omnibus gender inclusive term. There is some controversy whether this is entirely appropriate for use with both genders. That is, the etiology of the female triad being related to energy intake and availability (i.e., energy deficient) is well documented, but relative to EHMC this point is not entirely conclusive and remains to be substantiated [42]. Nonetheless, both conditions are associated with endocrine dysregulation of the HPA and hypothalamic-pituitary-gonadal axes.

Summary

Endurance training has the ability to affect a multitude of bodily systems. The effects manifest in changes in resting as well as in exercise responses for nearly all physiological parameters. The net impact of these changes is to improve the ability and efficiency of the body to perform prolonged aerobic activities. These adaptive changes span the continuum of organizational structure for humans—molecular, cellular, tissue, organ, organismal systems. To this end, the key endurance exercise adaptive changes are:

- ↑ Mitochondrial biogenesis
- ↑ Angiogenesis
- ↑ Fat oxidation
- ↑ $\text{VO}_{2\text{max}}$ (via ↑ CO and a-v O_2 difference)
- ↑ Ability to work at higher fractional utilization of $\text{VO}_{2\text{max}}$
- ↑ Work economy
- ↑ Musculotendinous resiliency
- ↑ Bone mineral density (i.e., weight bearing activities)
- ↑ Hormone—target tissue sensitivity
- ↑ Endocrine “down-stream” actions

It is important to recognize that these adaptations are not only critical to the endurance athlete, but equally important to improving the health and well-being of the general public. Regrettably, the vast majority of the research conducted on exercise adaptive responses has centered on men, with the assumption that male findings are directly transferable to females. In some parameters, this assumption is valid, but not all, and far more research is necessary to understand the adaptive responses of women to endurance exercise training [43].

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